Advanced Trade Studies and Decision Analysis Methods

The Analytic Hierarchy Process

Analytic Hierarchy Process

- Thomas Saaty, Univ. of Pennsylvania 1980.
- Software packages:
  - Expert Choice, MakeItRational, Triptych.
- Applied to a variety of decision analysis problems.
  - Transportation planning, portfolio selection, corporate planning, marketing, design trade studies, etc.
- Considers pairwise comparisons of elements (alternatives and attributes).
- Uses a scale indicating the strength with which one element dominates another with respect to a higher-level element.
  - Scaling process is translated into priority weights.

Analytic Hierarchy Process

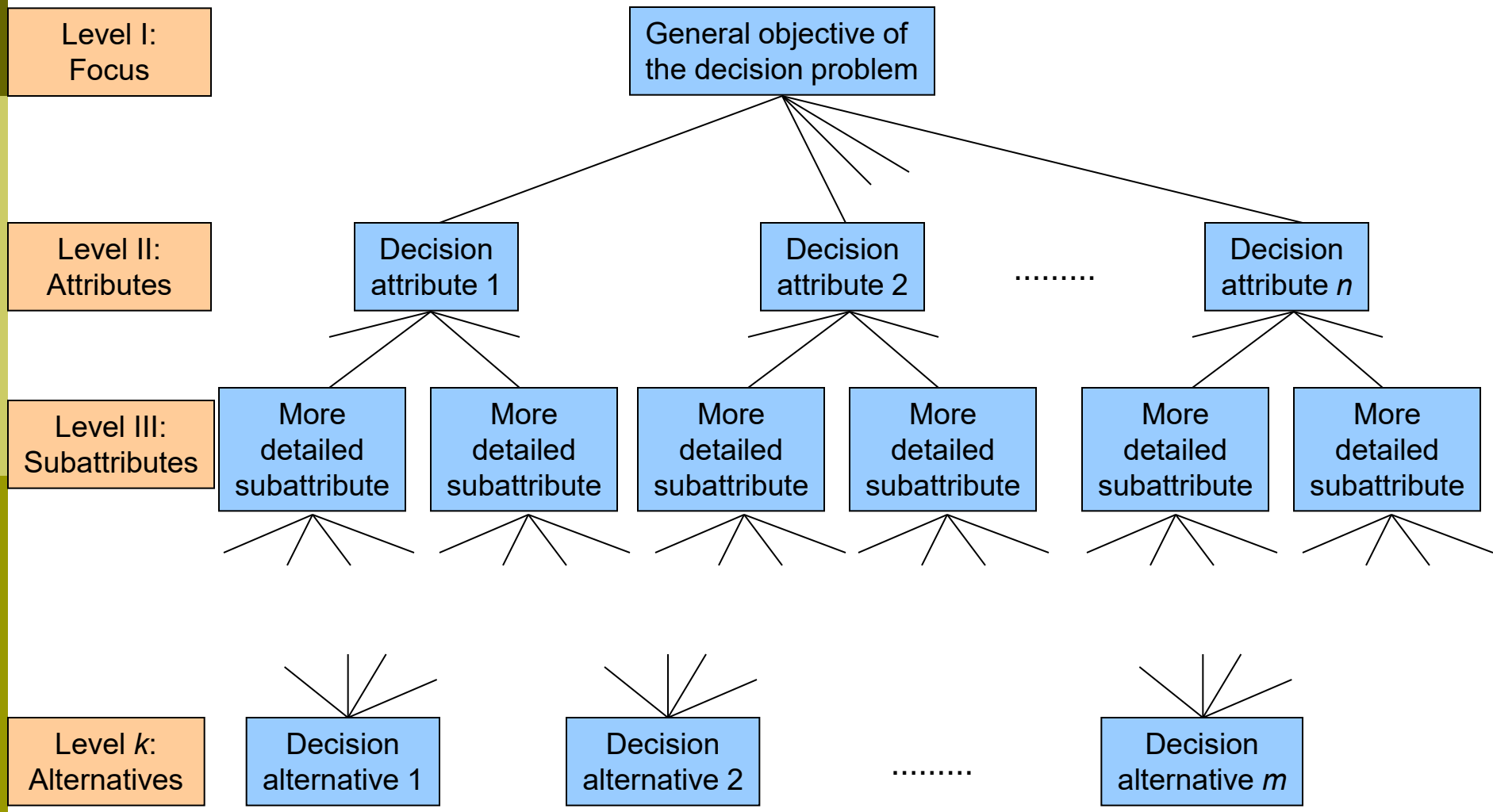
- Stages:
  - 1. Construct a decision hierarchy by breaking down the decision problem into a hierarchy of decision elements and identify decision alternatives.
  - 2. Determine the relative importance of attributes and sub-attributes, if any.
  - 3. Determine the relative standing (weight) of each alternative with respect to each next-higher level attributes or sub-attributes.
  - 4. Determine the indicator(s) of consistency in making pairwise comparisons.
  - 5. Determine the overall priority weight (score) of each alternative.

1. Construct Decision Hierarchy

- Decompose a complex problem into a hierarchy of sub-problems.

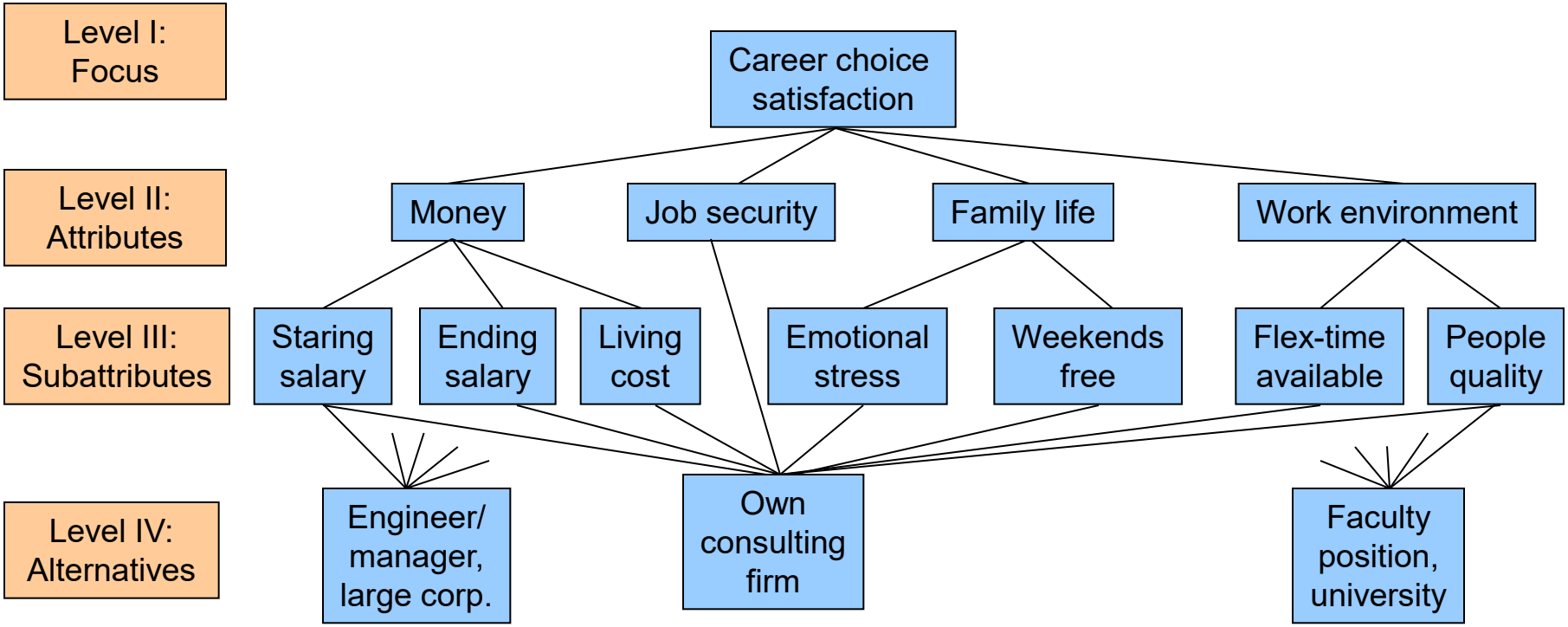
- Elements: overall objective, attributes, sub-attributes, sub-sub-attributes, alternatives.
- Top level is called focus: the broad overall objective.
- Subsequent levels may each have several elements, usually between 5 and 9.
- Elements from one level are compared with one another against a criterion in the next higher level.

1. Standard AHP Form



1. Principle of Hierarchic Composition

- Elements on a single level of the hierarchy are independent and their relative importance (priority weights) does not depend on the elements at the next lower level of hierarchy.



1. Construct Decision Hierarchy

- Other guidelines:
  - The number of levels used should be chosen to represent the problem effectively.
  - The order of the levels should reflect a logical causal relationship between adjacent levels.
  - The number of members in a particular level should be chosen to describe the level in adequate detail but should not cause unnecessary complexity.

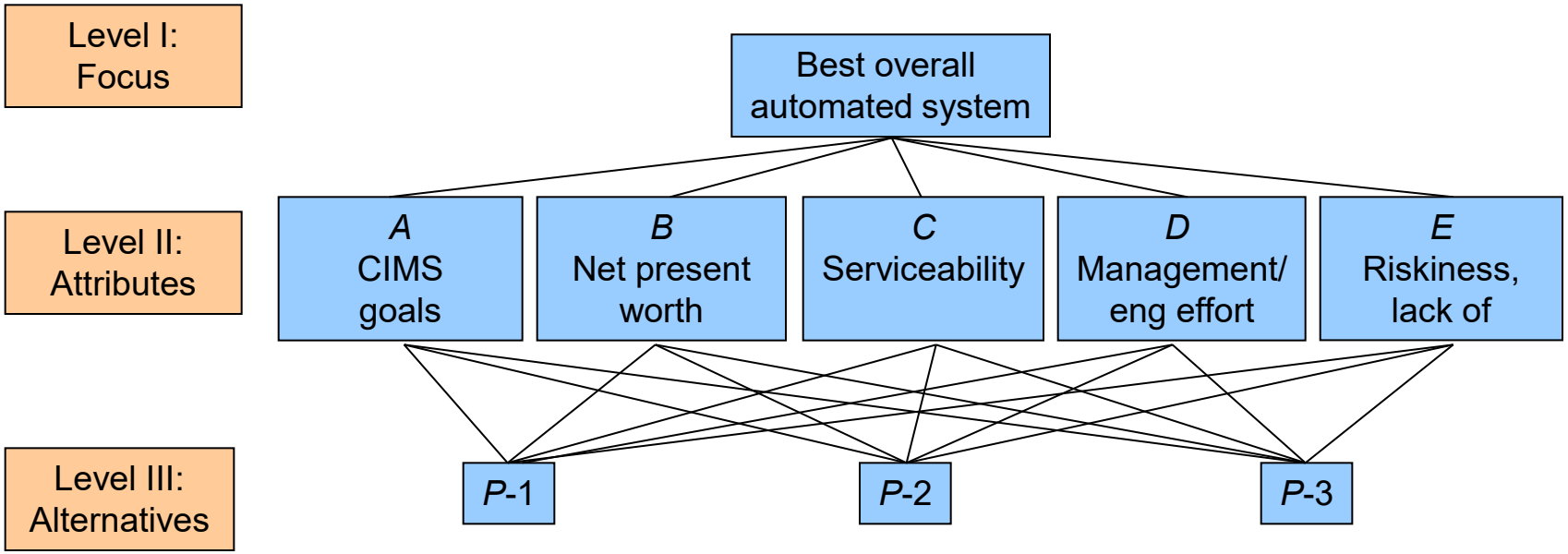
1. Construct Decision Hierarchy

- Example:
  - Qualitative and quantitative estimates are provided for the performance of 5 attributes used to judge the "value" or "worth" of four alternatives for modernizing a manufacturing company.

Attribute	**Alternative**			
	P-1	*P*-2	*P*-3	*P*-4
A. CIMS goals	Average	**Above average**	Average	Below average
B. Net present worth	$300M	$350M	**$400M**	$290M
C. Serviceability	**< 1 hr.**	1 - 1.5 hrs.	1 - 1.5 hrs.	1.5 - 2 hrs.
D. Management/ engineering effort	**500 hrs.**	700 hrs.	1,100 hrs.	700 hrs.
E. Riskiness, lack of	**Above average**	Average	Below average	Average

1. Construct Decision Hierarchy

- Example:
  - Drop *P*-4 because it is dominated by *P*-1 and *P*-2.



2. Determine Relative Importance of Attributes and Sub-attributes

- Priorities (relative importance weights) must be established for each set of elements at every level of the hierarchy.
  - Evaluate a set of elements at one level in a pairwise fashion regarding their relative importance with respect to an element at the next higher level of the hierarchy.
  - Answer to questions such as:
    - "With respect to the best overall system, how much more important is net present worth than lack of riskiness?"
    - "With respect to serviceability, how strongly do you prefer *P*-1 or *P*-2?"
  - Response is a value from 1 to 9 and its reciprocal, and the magnitude indicates the strength of the preference.

2. Determine Relative Importance of Attributes and Sub-attributes

- Degrees of preference.

If *x* is… as (than) *y*,	then the preference number to assign is
Equally important/preferred	1
Moderately important/preferred	3
Strongly important/preferred	5
Very strongly important/preferred	7
Extremely important/preferred	9

  - Even numbers (2, 4, 6, 8) can be used to represent compromises among the preferences above.

2. Determine Relative Importance of Attributes and Sub-attributes

- Major assumptions:
  - The relative weights of attributes within a level, conditional on each attribute in the immediately preceding level, are unidimensional.
  - Pairwise judgments of attributes encompass all relevant aspects of importance.
  - Decision makers can evaluate sub-problems in an accurate and relatively consistent manner.

2. Determine Relative Importance of Attributes and Sub-attributes

- Matrix of paired comparisons for attributes.

Attribute	*A*	*B*	*C*	*D*	*E*	Decimal equivalents				
						A	*B*	*C*	*D*	*E*
A. CIMS goals	1	1/3	5	6	5	1	0.333	5	6	5
B. Net present worth	3	1	6	7	6	3	1	6	7	6
C. Serviceability	1/5	1/6	1	3	1	0.20	0.167	1	3	1
D. Management / engineering effort	1/6	1/7	1/3	1	1/4	0.167	0.143	0.333	1	0.25
E. Lack of riskiness	1/5	1/6	1	4	1	0.20	0.167	1	4	1
∑						4.567	1.810	13.333	21	13.25

2. Determine Relative Importance of Attributes and Sub-attributes

- Compute the vector of priorities.
  - Normalized matrix of paired comparisons and calculations of priority weights.

	A	***B***	***C***	***D***	***E***	**Row Σ**	**Average = Σ /5**
A	0.219	0.184	0.375	0.286	0.377	1.441	0.288
B	0.657	0.553	0.450	0.333	0.453	2.446	0.489
C	0.044	0.092	0.075	0.143	0.076	0.430	0.086
D	0.036	0.079	0.025	0.048	0.019	0.207	0.041
E	0.044	0.092	0.075	0.190	0.074	0.477	0.096
	Σ=1.000	Σ=1.000	Σ=1.000	Σ=1.000	Σ=1.000	S = 5.000	Σ=1.000

3. Determine the Relative Standing of Alternative with Respect to Attributes

- Determine the principal vector (priorities) of each alternative with respect to each of the attributes to which they relate in the next higher level.
  - Use subjective judgments to prioritize alternatives with respect to attributes.
  - Use performance data to prioritize alternatives with respect to attributes.

3. Determine the Relative Standing of Alternative with Respect to Attributes

- Using subjective judgments to prioritize alternatives with respect to attributes.
  - Matrix of paired comparison results for alternatives with respect to the first attribute (*A*).

	***P*-1**	***P*-2**	***P*-3**	***P*-1**	***P*-2**	***P*-3**
P-1	1	1/3	1	1	0.333	1
P-2	3	1	2	3	1	2
P-3	1	1/2	1	1	0.500	1
				Σ=5	Σ=1.833	Σ=4

  - Normalized matrix and priority weights for alternatives with respect to the first attribute (*A*).

	***P*-1**	***P*-2**	***P*-3**	**Σ**	**Approx. principal vector (Σ/3)**
P-1	0.20	0.18	0.25	0.63	0.21
P-2	0.60	0.55	0.50	1.65	0.55
P-3	0.20	0.27	0.25	0.72	0.24
	Σ=1.00	Σ=1.00	Σ=1.00		Σ=1.00

3. Determine the Relative Standing of Alternative with Respect to Attributes

- Summary of paired comparisons and resulting priority weights for alternatives with respect to selected attributes.

		P-1	*P*-2	*P*-3	Priority weights
A. CIMS Goals	*P*-1	1	1/3	1	0.21
	P-2	3	1	2	0.55
	P-3	1	1/2	1	0.24
					Σ=1.00
C. Serviceability	*P*-1	1	2	2	0.50
	P-2	1/2	1	1	0.25
	P-3	1/2	1	1	0.25
					Σ=1.00
E. Riskiness, lack of	*P*-1	1	3	4	0.62
	P-2	1/3	1	2	0.24
	P-3	1/4	1/2	1	0.14
					Σ=1.00

3. Determine the Relative Standing of Alternative with Respect to Attributes

- Using performance data to prioritize alternatives with respect to attributes.
  - Rate alternatives using performance data (when available) instead of pairwise comparison process.

Attribute	Unit of measure	Alternative		
		P-1	*P*-2	*P*-3
B. Net present worth	$	300M	350M	400M
D. Management/eng effort	Hours required	500	700	1100

P-1: $300M/($300M+$350M+$400M)=0.29

P-2: $350M/($300M+$350M+$400M)=0.33

P-3: $400M/($300M+$350M+$400M)=0.38

$\sum = 1.00$

3. Determine the Relative Standing of Alternative with Respect to Attributes

- Priority weights for alternatives with respect to quantified attributes.

Alternative	*B*. Net present worth	*D*. Management / eng effort
P-1	0.29	0.46
P-2	0.33	0.33
P-3	0.38	0.21
	Σ=1.00	Σ=1.00

4. Determine the Consistency of Making Pairwise Comparisons

- Measure the degree of consistency present in the subjective judgments.
- Perfect cardinal consistency.
  - If attribute *A* is judged to be twice as important as attribute *B*, and attribute *B* is judged to be twice as important as attribute *C*, then attribute *A* would be judged to be four times as important as attribute *C*.
- Method to measure the magnitude of departure from perfect consistency.
  - Local measure of consistency for individual comparison matrices.
    - Local consistency ratio (*C.R.*) is an approximate mathematical indicator of the consistency of pairwise comparisons.
  - Global measure of consistency for the entire decision problem.

4. Local Measure of Consistency

- Local consistency ratio (*C.R.*) is an approximate mathematical indicator of the consistency of pairwise comparisons.
- *C.R.* = *CI*/*RI*
  - *CI* = consistency index: function of the vector of priorities and the size of the matrix.
  - *RI* = random index: if the pairwise comparisons were made at random.
- *C.R.* ≤ 0.1, consistency is acceptable.

4. Local Measure of Consistency

- Example
  - Multiply the matrix of comparisons [*A*] by the principal vector or priority weights [*B*] to get vector [*C*]

$$
\begin{array}{ccccc}
 & & [A] & & \\
\end{array}
\qquad
[B] \qquad [C]
$$

$$
\begin{vmatrix} 1 & 0.33 & 5 & 6 & 5 \\ 3 & 1 & 6 & 7 & 6 \\ 0.2 & 0.167 & 1 & 3 & 1 \\ 0.167 & 0.143 & 0.333 & 1 & 0.25 \\ 0.2 & 0.167 & 1 & 4 & 1 \end{vmatrix}
\times
\begin{vmatrix} 0.288 \\ 0.489 \\ 0.086 \\ 0.041 \\ 0.096 \end{vmatrix}
=
\begin{vmatrix} 1.607 \\ 2.732 \\ 0.444 \\ 0.212 \\ 0.485 \end{vmatrix}
$$

  - Divide each element in vector [*C*] by its corresponding element in vector [*B*] to find a new vector [*D*]

$$
[D] = \begin{vmatrix} \frac{1.607}{0.288} & \frac{2.732}{0.489} & \frac{0.444}{0.086} & \frac{0.212}{0.041} & \frac{0.485}{0.096} \end{vmatrix}
$$

$$
= \begin{vmatrix} 5.58 & 5.59 & 5.16 & 5.17 & 5.05 \end{vmatrix}
$$

4. Local Measure of Consistency

- Example
  - Average the numbers in vector [*D*] to obtain λ_{max}
  - λ_{max} =(5.58+5.59+5.16+5.17+5.05)/5 = 5.31
  - $CI = (\lambda_{max} - N)/(N-1) = (5.31-5)/(5-1) = 0.08$
  - *RI* for various matrix sizes have been approximated based on large numbers of simulation runs as follows:

N	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**…**
RI	0.00	0.00	0.58	0.90	1.12	1.24	1.32	1.41	1.45	1.49	1.51	…

  - $C.R. = CI/RI = 0.08/1.12 = 0.07 < 0.10$ – acceptable

4. Local Measure of Consistency

- Example:
  - Calculation for alternative comparisons with respect to CIMS goals.

$$\begin{array}{c} [A] \\ \begin{vmatrix} 1 & 0.333 & 1 \\ 3 & 1 & 2 \\ 1 & 0.5 & 1 \end{vmatrix} \end{array} \times \begin{array}{c} [B] \\ \begin{vmatrix} 0.21 \\ 0.55 \\ 0.24 \end{vmatrix} \end{array} = \begin{array}{c} [C] \\ \begin{vmatrix} 1.633 \\ 1.660 \\ 0.725 \end{vmatrix} \end{array}$$

$$[D] = \begin{vmatrix} \frac{0.633}{0.21} & \frac{1.660}{0.55} & \frac{0.725}{0.24} \end{vmatrix} = \begin{vmatrix} 3.01 & 3.02 & 3.02 \end{vmatrix}$$

λ_{max} =(3.01+3.02+3.02)/3 = 3.02

  - CI = (3.02-3)/(3-1) = 0.01
  - $C.R.$ = CI/RI = 0.01/0.58 = 0.02 < 0.10 – acceptable

4. Local Measure of Consistency

- Example:
  - Consistency indices and ratios for the comparison matrices of alternatives with respect to attributes *A*, *C*, *E*.

	Consistency index, *CI*	**Consistency ratio, *C.R.***
A. CIMS goals	0.01	0.02
C. Serviceability	0.00	0.00
E. Lack of riskiness	0.01	0.02

  - Same approach for *B* and *D*.

5. Determine the Overall Priority Weights of Alternatives

- Weighted evaluation is obtained by multiplying the matrix of evaluation ratings by the vector of priority weights and summing over all attributes.

  Weighted evaluation of alternative k =

  Σ (priority weights$_i$ x evaluation rating$_{ik}$)

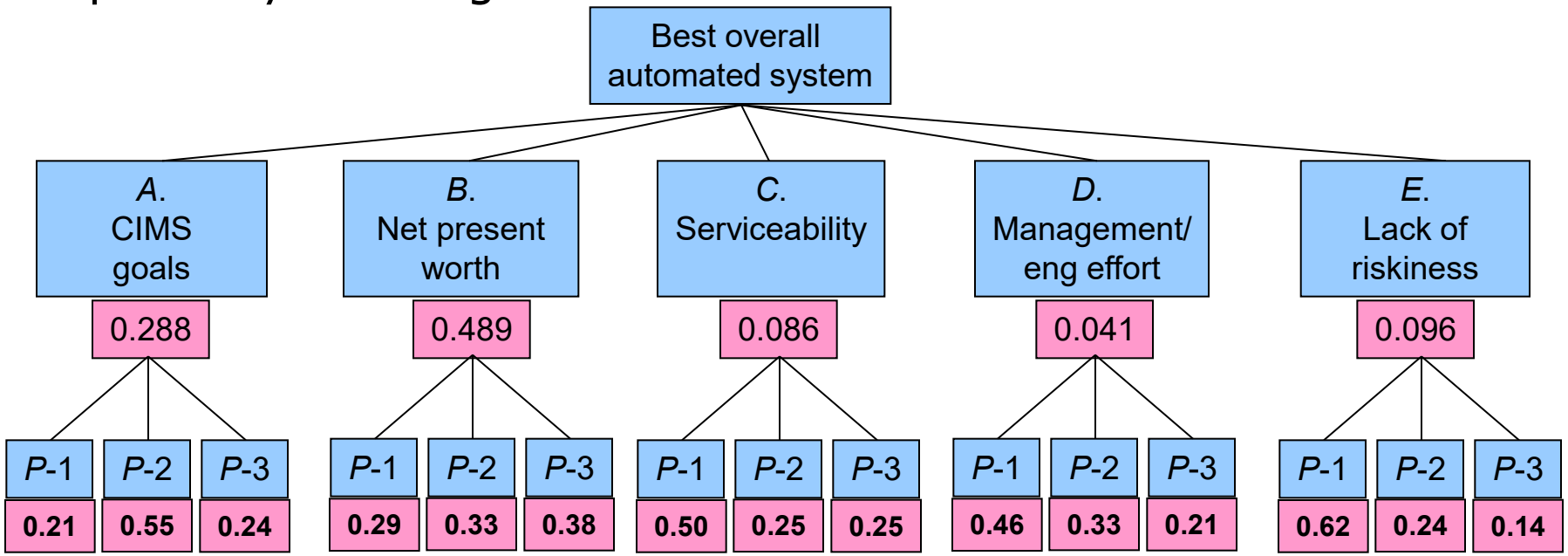
	Attribute					
Attribute weights	*A*	*B*	*C*	*D*	*E*	Alternative priority weights
	0.288	0.489	0.086	0.041	0.096	
Alternative						
P-1	0.21	0.29	0.50	0.46	0.62	0.324
P-2	0.55	0.33	0.25	0.33	0.24	0.378
P-3	0.24	0.38	0.25	0.21	0.14	0.298

Priority weight for alternative *P*-1 is:

0.288(0.21)+0.489(0.29)+0.086(0.5)+0.041(0.46)+0.096(0.62)=0.324

5. Determine the Overall Priority Weights of Alternatives

- Calculate priority weight (weighted evaluation) for any alternative by summing the products of weights for all pathways leading to that alternative.



- For alternative *P*-1

  0.288(0.21)+0.489(0.29)+0.086(0.5)+0.041(0.46)+0.096(0.62)=0.324

5. Global Consistency Ratio: *C.R.H.*

- Measurement of consistency is applied to the entire decision hierarchy.
- *C.R.H.* is obtained by taking the ratio of an aggregate consistency index M_{CI} for the entire three-level hierarchy to an aggregate random index M_{RI}.
- M_{CI} = second-level *CI* + | vector of second-level priority weights | x | vector of third-level *CI*s |
- M_{RI} = second-level *RI* + | vector of second-level priority weights | x | vector of third-level *RI*s |
- If $C.R.H. = M_{CI}/M_{RI} \leq 0.1$, the consistency of the hierarchy is acceptable.

Global Consistency Ratio: *C.R.H.*

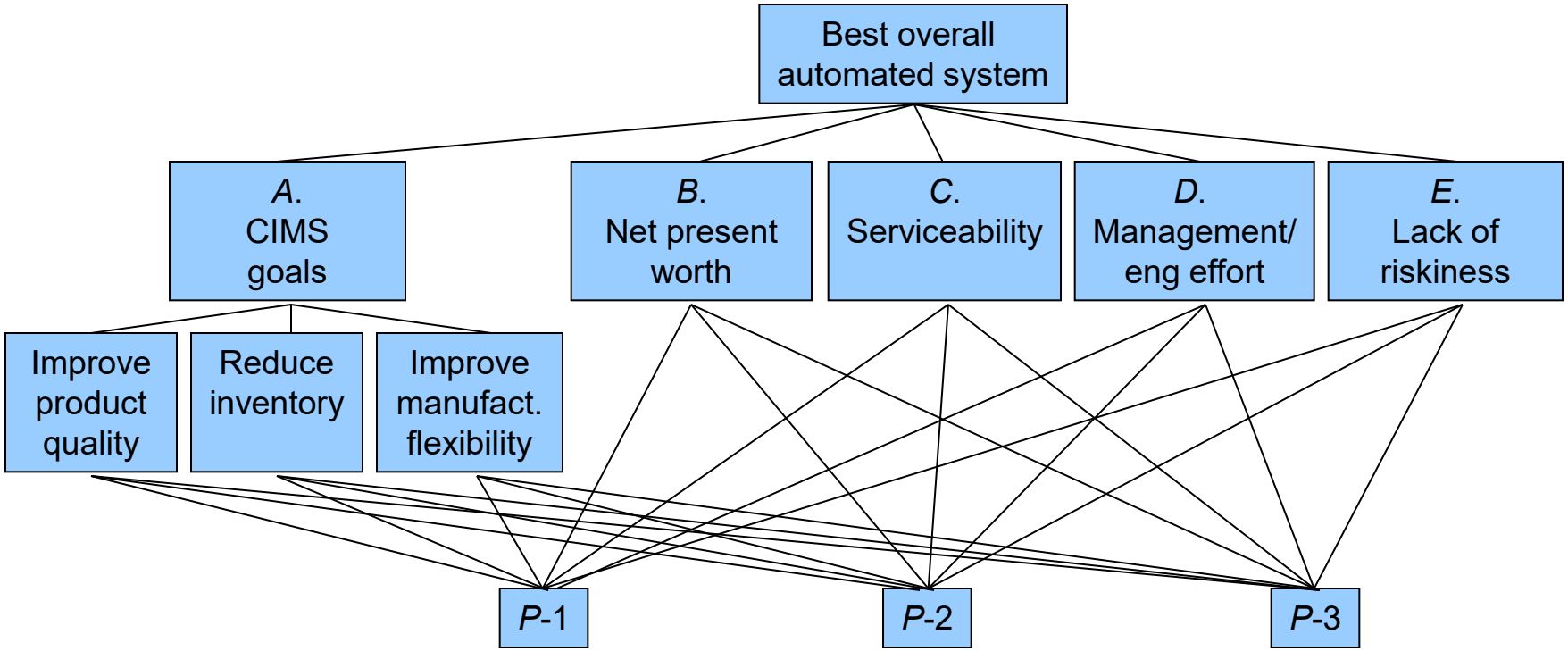
- Example
- $M_{CI} = 0.08 + \begin{vmatrix} 0.288 & 0.489 & 0.086 & 0.041 & 0.096 \end{vmatrix} \times \begin{vmatrix} 0.01 \\ 0.00 \\ 0.00 \\ 0.00 \\ 0.01 \end{vmatrix} = 0.08$
- $M_{RI} = 1.12 + \begin{vmatrix} 0.288 & 0.489 & 0.086 & 0.041 & 0.096 \end{vmatrix} \times \begin{vmatrix} 0.58 \\ 0.58 \\ 0.58 \\ 0.58 \\ 0.58 \end{vmatrix} = 1.7$
- $C.R.H. = M_{CI}/M_{RI} = 0.08/0.7 = 0.05 < 0.1$ - acceptable.

What if there is Inconsistency

- Decision maker should review the judgments made and look for inconsistency.
- Rank ordering the elements in the matrix highlights inconsistencies.
  - Ratings in the upper diagonal of the matrix should be greater than or equal to 1 and increase in value across the rows of the matrix.
- If the source of inconsistency is not apparent or cannot be resolved satisfactorily through the re-estimation of preferences, it is possible that there is a problem with the hierarchical formulation.
  - The attributes or alternatives being compared are not independent or are not directly comparable on 1-9 scale.

Hierarchies with More than 3 Levels

- Four-level hierarchy.
- Example: consider 3 subattributes for the second-level attribute CIMS goals.



Determine the Relative Weights of Attributes and Alternatives

- Matrix of paired comparisons and priority weights for attributes.

With respect to the "best overall automated system"	*A*	*B*	*C*	*D*	*E*	Priority weight	Consistency
A. CIMS goals	1	1/3	5	6	5	0.288	
B. Net present worth	3	1	6	7	6	0.489	
C. Serviceability	1/5	1/6	1	3	1	0.086	*CI* = 0.08
D. Management/eng effort	1/6	1/7	1/3	1	1/4	0.041	*C.R.* = 0.07
E. Lack of riskiness	1/5	1/6	1	4	1	0.096	
						Σ=1.000	

Determine the Relative Weights of Attributes and Alternatives

- Matrix of paired comparisons and priority weights for sub-attributes.

With respect to the "CIMS goals"	*A*1	*A*2	*A*3	Priority weight	Consistency
*A*1. CIMS goals	1	2	5	0.59	*CI* = 0.003
*A*2. Net present worth	1/2	1	2	0.28	
*A*3. Serviceability	1/5	1/2	1	0.13	*C.R.* = 0.005
				Σ=1.000	

Determine the Relative Weights of Attributes and Alternatives

- Priority weights of alternatives with respect to selected attributes.

	Attribute			
	B	*C*	*D*	*E*
P-1	0.29	0.50	0.46	0.62
P-2	0.33	0.25	0.33	0.24
P-3	0.38	0.25	0.21	0.14
CI	0.00	0.00	0.00	0.01
C.R.	0.00	0.00	0.00	0.02

Determine the Relative Weights of Attributes and Alternatives

- Summary of paired comparisons and resulting priority weights for alternatives with respect to sub-attributes.

		P-1	*P*-2	*P*-3	Priority weights
*A*1. Improve product quality	*P*-1	1	1/2	1	0.26
	P-2	2	1	1/9	0.21
	P-3	1	9	1	0.53
*A*2. Reduce inventory	*P*-1	1	1/2	1	0.24
	P-2	2	1	3	0.55
	P-3	1	1/3	1	0.21
					Σ=1.00
*A*3. Improve manufacturing flexibility	*P*-1	1	1/8	1/3	0.08
	P-2	8	1	3	0.68
	P-3	3	1/3	1	0.24
					Σ=1.00

Local Measure of Consistency

- Summary of consistency index and consistency ratio.

	Consistency index, *CI*	**Consistency ratio, *C.R.***
*A*1. Improve product quality	0.550	0.948
*A*2. Reduce inventory	0.009	0.016
*A*3. Improve manufacturing flexibility	0.001	0.002

Determine the Relative Weights of Attributes and Alternatives

- *C.R.* for improved product quality = 0.948 > 0.10.
  - Judgment made were inconsistent.
  - *P*-2 is twice as preferred as *P*-1.
  - *P*-1 is equally preferred to *P*-3.
  - So, *P*-2 should be twice as preferred as *P*-3.
  - However, *P*-3 has been judged to be 9 times more preferred than *P*-2.
- Solving the inconsistency.
  - Assume *P*-2 is twice as preferred as both *P*-1 and *P*-3.

Determine the Consistency of the Comparisons

- Revised paired comparisons and resulting priority weights for alternatives with respect to sub-attributes.

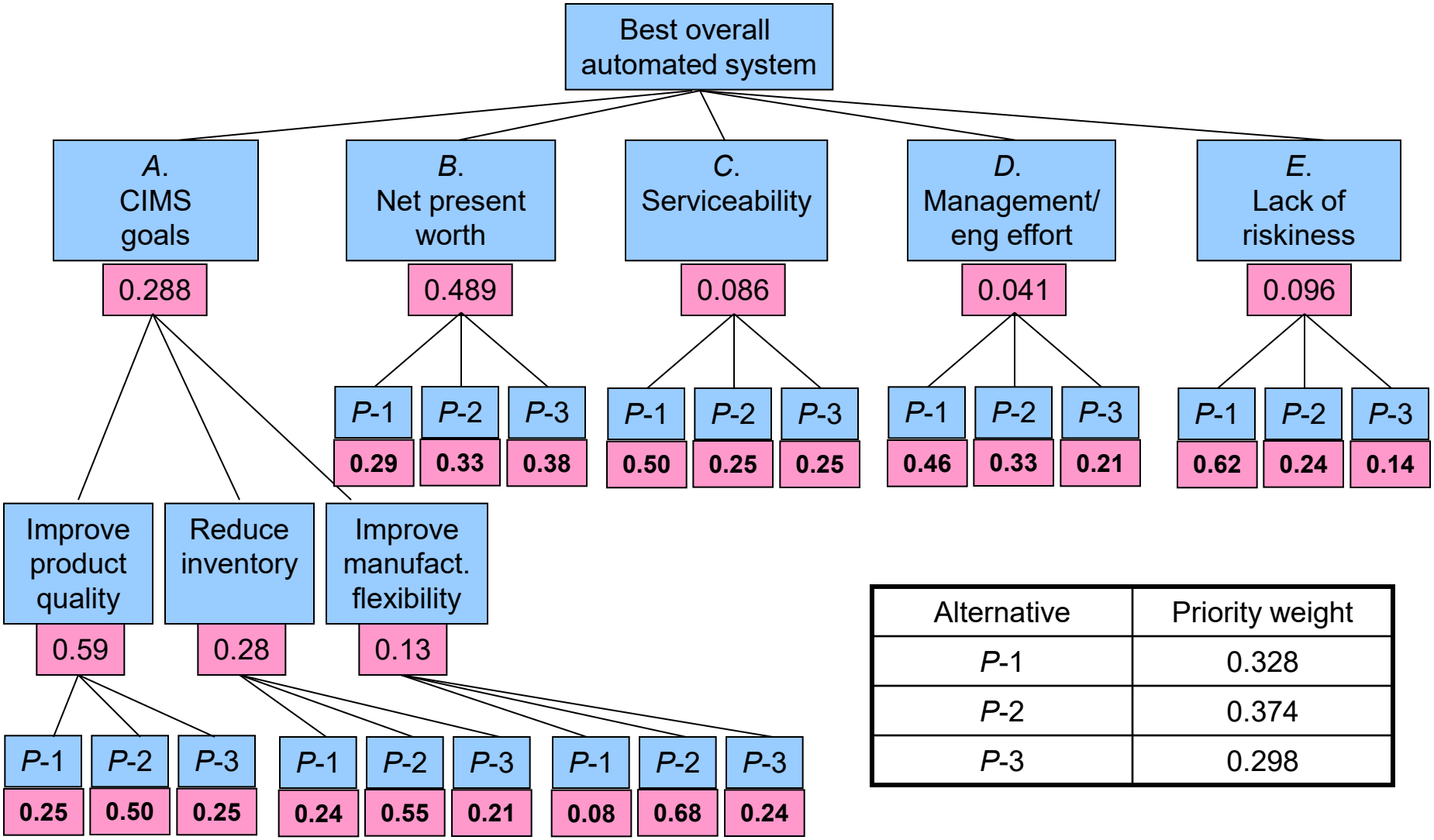
		P-1	*P*-2	*P*-3	Priority weights
*A*1. Improve product quality	*P*-1	1	1/2	1	0.25
	P-2	2	1	1/9	0.50
	P-3	1	9	1	0.25
					Σ=1.00
*A*2. Reduce inventory	*P*-1	1	1/2	1	0.24
	P-2	2	1	3	0.55
	P-3	1	1/3	1	0.21
					Σ=1.00
*A*3. Improve manufacturing flexibility	*P*-1	1	1/8	1/3	0.08
	P-2	8	1	3	0.68
	P-3	3	1/3	1	0.24
					Σ=1.00

Determine the Consistency of the Comparisons

- Summary of revised consistency index and consistency ratio.

	Consistency index, *CI*	**Consistency ratio, *C.R.***
*A*1. Improve product quality	0.000	0.000
*A*2. Reduce inventory	0.009	0.016
*A*3. Improve manufacturing flexibility	0.001	0.002

Determine the Overall Weights of Alternatives



Alternative	Priority weight
P-1	0.328
P-2	0.374
P-3	0.298

If the Inconsistency Were Unsolved

- If priority weights from first table were used:

Alternative	Priority weight
P-1	0.330
P-2	0.325
P-3	0.345

  - *P*-3 more desirable than *P*-1 or *P*-2, which is wrong.

Global Consistency Measure

- Same procedure as before considering a three-level hierarchy composed of CIMS goals, its sub-attributes, and the alternatives.
- Obtain an aggregate consistency index M_{CI} for the CIMS goals attribute.
  - $M_{CI} = 0.006$
- Use the obtained value in conjunction with the other second-level attributes to obtain the *C.R.H.* for the entire decision problem.
  - $M_{CI} = 0.083$
  - $M_{RI} = 1.7$
  - $C.R.H. = M_{CI}/M_{RI} = 0.049$ – acceptable.

Issues in Applying AHP

- Vagueness of the questioning procedure.
  - Decision maker must think of some average quantity when comparisons are meaningless.
- Rank reversal.
  - Introduction of a new alternative reverses the rankings of previously evaluated alternatives.
  - Decision attributes should be independent of each other
  - Attributes should be independent of the alternatives being considered.
- Aggregating benefits and costs.
  - When benefit hierarchy and cost hierarchy are used, the highest benefit-cost ratio alternative is used to find the best alternative, but it does not yield optimal results.
  - Correct analysis needs to consider incremental benefits and costs.

Quiz – Q1

- What is the primary method used in the Analytic Hierarchy Process (AHP) to evaluate elements within the decision hierarchy?
  A. Aggregating expert opinions through a voting system.
  B. Direct assignment of scores on a 1-100 scale.
  C. Pairwise comparisons of elements.
  D. Calculating the net present worth for all alternatives.

Quiz – Q2

- In the structure of an AHP decision hierarchy, what is the top level that represents the broad overall objective called?
  A. Focus
  B. Attributes
  C. Alternatives
  D. Objective

Quiz – Q3

- When conducting pairwise comparisons in AHP using the 1-9 scale, what does a preference number of '1' signify when comparing element A to element B?
  A. Element A is weakly more important than Element B.
  B. The comparison is inconsistent or invalid.
  C. Element A is absolutely more important than Element B.
  D. Element A and Element B are equally important/preferred.

Quiz – Q4

- A key step in AHP is to check the consistency of subjective judgments. What is the generally accepted threshold for the local Consistency Ratio (C.R.)?
  A. The C.R. should be less than or equal to 0.10.
  B. The C.R. should be exactly 0.
  C. The C.R. should be less than or equal to 1.0.
  D. The C.R. should be greater than the Random Index (RI).

Quiz – Q5

- The formula for Consistency Index (CI) is $CI = \frac{\lambda_{max} - N}{N-1}$. What does the term 'N' represent in this formula?
  A. The size of the comparison matrix
  B. The number of levels in the hierarchy
  C. The normalized priority weight
  D. The Random Index (RI) value

Quiz – Q6

- How is the final overall priority weight, or score, for a specific alternative determined in AHP?
  A. By averaging the alternative's priority weights across all attributes.
  B. By summing the products of each attribute's weight and the alternative's corresponding priority weight for that attribute.
  C. By selecting the attribute under which the alternative has the highest priority weight.
  D. By multiplying all the priority weights along the paths leading to the alternative.

Quiz – Q7

- When evaluating alternatives against an attribute with quantitative data (e.g., Net Present Worth of $300M, $350M, $400M), how can priority weights be derived without subjective pairwise comparisons?
  A. By normalizing the performance data, for instance, by dividing each alternative's value by the sum of all values.
  B. By assigning pairwise comparison scores based on the ratios of the data (e.g., 400/300).
  C. By ranking the alternatives from best to worst and assigning fixed weights.
  D. The AHP process requires that all evaluations must use the 1-9 subjective scale.

Quiz – Q8

- The source material identifies 'rank reversal' as a potential issue when applying AHP. What is rank reversal?
  A. An alternative's final score is lower than its score on the most important attribute.
  B. The priority weights of the attributes change when a new decision maker provides input.
  C. The relative rankings of existing alternatives change when a new alternative is introduced.
  D. A decision maker reverses their own preference during a second round of comparisons.

Quiz – Q9

- If a pairwise comparison matrix is found to be highly inconsistent (e.g., a C.R. value significantly greater than 0.1), what is the first recommended course of action?
  A. Accept the inconsistency as long as the global consistency ratio (C.R.H) is acceptable.
  B. Remove the attribute from the decision hierarchy.
  C. Review the subjective judgments to find and re-estimate the source of the inconsistency.
  D. Average the inconsistent judgments to smooth out the data.

Homework posted